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MAIIC IFRS 9 and EIR system

MAIIC Staging Module Specification v1

Governed staging and the staging comparison run

15 October 2026

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For Dr Thomson Kumwenda (CFO), MAIIC Finance, Credit and ICT; companion to the EIR Engine Specification v4 (section 3.6) and the FLI Module Specification v1

Draft for review

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1. What this specification does

Staging decides whether each loan is in Stage 1, 2 or 3 under IFRS 9 at every month-end. It drives more of the system than any other single rule. The probability of default is measured on stage movements, the expected credit loss is twelve-month or lifetime by stage, interest on a Stage 3 loan is recognised on the net basis, and the Mega Farm provision, the restructure register, the forward-looking route and the IFRS 9 note all read the stage.

Today the staging rules sit in the code. This specification makes them **governed**: each rule becomes a setting in the Governance Centre, proposed by one person and approved by another, with an effective date and a comment, like every other policy setting in the system. It makes every change in the Governance Centre fully logged, with room for supporting documents (section 8). It also adds a **staging comparison run**, which stages any month under a second set of rules beside the booked one and explains every difference, without changing the booked figures.

Two sets of rules are defined:

- **Single-ladder staging**: MAIIC's staging policy as at 31 December 2025, as applied in the ECL workbooks of 14 December 2025. It is the **seeded default for every period**.
- **Directive-aligned staging**: the rules the system applies today, built on the Reserve Bank of Malawi's directive for development finance institutions and MAIIC's published accounting policy. It is **defined but not in force**. It takes effect only when MAIIC adopts it, through a governed proposal and approval with an effective date chosen by MAIIC.

2. Why

2.1 Three bases that do not agree

Three staging bases are in play, and they differ:

Point	RBM DFI Directive 2018	ECL workbooks of December 2025	System today
When a loan becomes non-performing (Stage 3)	Short-term 91 days; medium and long-term 181 days	181 days for every facility	By term, as the directive: 91 and 181 days; Mega Farm 91
Stage 2	Special mention from 31 days (short-term) or 91 days (medium and long-term)	31 days for every facility	31 days for every facility
How days past due are counted	Days the instalments are overdue	The highest arrears bucket in the loan book report	From the oldest overdue instalment, with a count from maturity and corrections against the instalment schedule in force
Missed-instalment default (note 22.8.1)	Not addressed	Not applied	4 consecutive instalments
Qualitative triggers	Not addressed	None	Approved triggers on the SICR register
Leaving a stage	Back to accrual only when all arrears are paid (s.13(3))	Re-staged afresh at each year-end	Held for 3 month-ends below the threshold
Frequency	Quarterly review (s.11)	Annual	Every month-end

At 31 December 2025, the workbooks put the main book at 80, 10 and 18 facilities in Stages 1, 2 and 3. The system on 14 October 2026 put it at 50, 22 and 51. Neither figure can be explained without separating the rules from the data, which is what the comparison run does.

2.2 Why the 31 December 2025 policy is the default

MAIIC has not adopted a new staging policy. The system should therefore stage the way MAIIC staged until MAIIC decides otherwise. The improved rules are fully built and available, and the comparison run shows their effect every month, so the decision can be taken on evidence rather than on argument.

2.3 What this choice costs until the new policy is adopted

With the 31 December 2025 policy in force, the system will knowingly depart from the directive and from MAIIC's published policy in the ways below. Each is recorded in the Governance Centre against the setting that causes it, and each is measured by the comparison run every month:

1. **Short-term facilities 91 to 180 days overdue stay in Stage 2.** The directive classes them substandard and IFRS 9 presumes default at 90 days; a short-term facility has no rebuttal.
2. **Matured loans with an unpaid balance can read as current.** Bucket ageing takes E-Banker's report as it stands, and that report leaves the arrears empty on many matured loans. On the system's data, the Mega Farm bullet loans that matured in 2025 read 0 days on about 47 billion kwacha. The December workbooks had them in Stage 3 only because their own loan book had the arrears filled in.
3. **The missed-instalment default is not applied**, although MAIIC's 2025 financial statements (note 22.8.1) describe it as the policy.
4. **Two proven E-Banker report defects are left uncorrected**: arrears aged from deleted instalment definitions, and collapsed schedules that show a paid-up loan as wholly overdue.
5. **A loan leaves its stage on its first good month-end**, with no probation.

These are accepted departures, not errors, for as long as MAIIC keeps the 31 December 2025 policy. They are the reasons to adopt the directive-aligned rules, and the comparison run measures what each one is worth.

3. The settings

Each setting below is a Governance Centre setting with options, a default, a description and a comment. The column "31 December 2025 policy" shows the option that reproduces the ECL workbooks of December 2025; its comment in the Governance Centre reads "*MAIIC's staging policy as at 31 December 2025 (ECL workbooks of 14 December 2025).*"

#	Setting	Options	31 December 2025 policy (seeded default)	Directive-aligned staging	Comment
1	`staging_threshold_profile`: where Stage 3 starts	One threshold for every facility (181 days) · By facility term (short-term 91, medium and long-term 181) · IFRS 9 presumption for every facility (91)	One threshold for every facility	By facility term	The directive classes facilities by repayment period, and 181 days for medium and long-term loans rests on its rebuttal of IFRS 9's 90-day presumption. The IFRS 9 option is kept as a conservative benchmark
2	`stage2_threshold_medium_long`: where Stage 2 starts for medium and long-term facilities	31 days · 91 days, with documented rebuttal evidence	31 days	31 days	Both policies keep IFRS 9's 30-day presumption. The 91-day option needs Dr Thom's decision O13 and documented evidence that late payment does not signal higher risk for this book
3	`staging_mega_farm_term`: how Mega Farm loans are classed	Same threshold as every facility · By tenor · Always short-term (91)	Same threshold as every facility	By tenor	The directive classes by repayment period, and some Mega Farm loans run five years. "Always short-term" rests on the seasonal-finance argument, which Finance has not confirmed (spec v4, section 16)
4	`dpd_basis`: how staging counts days past due	Bucket ageing of the overdue amounts · Oldest overdue instalment	Bucket ageing	Oldest overdue instalment	Exists today. Buckets cannot tell 200 days from 900. **This setting governs staging only; the RBM return always uses the directive's count (section 5)**
5	`dpd_after_maturity`: a matured loan still owing	Core-system ageing only · Count from maturity when longer	Core-system ageing only	Count from maturity	Without the maturity count, matured bullet loans can read current (section 2.3, item 2)
6	`dpd_schedule_correction`: when the loan book report contradicts the schedule in force	Core-system report as given · Recount against the schedule in force and the ledger	Report as given	Recount	Corrects two proven E-Banker defects; every corrected row is flagged with the reason
7	`stage3_missed_instalments`: default on missed payments	Not applied · 3 · 4 · 6 consecutive	Not applied	4 consecutive	Exists today. MAIIC's published policy (2025 statements, note 22.8.1) is 4

#	Setting	Options	31 December 2025 policy (seeded default)	Directive-aligned staging	Comment
8	`sicr_qualitative_register`: approved qualitative triggers	Not applied · Applied	Not applied	Applied	IFRS 9 B5.5.17 expects qualitative evidence beyond days past due. Only triggers approved on the register count
9	`sicr_pd_change`: the change in the probability of default since origination	Not applied · PD at least doubled since origination	Not applied	Not applied	The workbook template carries a PD-doubling test, but it never fires (the prior PD equals the current PD). Needs each loan's PD at origination, which the system does not yet keep. A later build
10	`stage_cure_months`: month-ends held before moving down	0 · 3 · 6 · 12	0	3	Exists today, seeded at 3; the seeded value changes to 0 under this specification
11	`stage_cure_test`: what counts as cured	Months below the threshold · All arrears cleared · Arrears cleared, then the months	Months below the threshold (no effect at 0 months)	Arrears cleared, then the months	The strictest option meets the directive's test for returning to accrual (s.13(3)) and IFRS 9's need for sustained performance
12	`staging_preset`: one choice that sets settings 1 to 11	Single-ladder staging · Directive-aligned staging · Custom	Single-ladder staging	Directive-aligned staging	A change of preset is one governed change; any later change to a single setting turns the preset to Custom and is recorded with its reason

3.1 How the defaults are seeded

- Every setting is seeded with the **31 December 2025 policy**, effective from inception (`1900-01-01`, as the other governed defaults), labelled as the seeded default and not as a MAIIC approval.
- **Directive-aligned staging is seeded as a defined preset only.** Nothing switches it on by date. It takes effect only through a proposal and a second person's approval in the Governance Centre, with the effective date MAIIC chooses. The bootstrap never approves it.
- The threshold rows on the Staging & SICR Rules screen stay as they are and are read through setting 1: the profile chooses which rows apply.

3.2 Adopting directive-aligned staging

When MAIIC decides to adopt it:

1. A person proposes the **Directive-aligned staging** preset with an effective date (normally the first day of a month not yet locked) and the reason, citing the board or committee decision.
2. A second person approves it.

3 From that month the booked staging follows the new rules. Locked months are never restated. Under IAS 8 the change is a change in accounting estimate, applied prospectively.

4 The comparison run for the first month under the new policy is kept as the record of its effect.

4. The staging comparison run

The comparison run answers one question: *what would this month's stages and ECL be under another set of rules, and why?*

What it does. For a chosen month and a chosen preset (or a custom set of settings), it stages every loan under those rules into a separate table, beside the booked stage. It never writes to the loan book, the ECL, the revenue or any report, and it is allowed in a locked month.

What it reports, per loan and in total:

Column	Meaning
Booked stage, comparison stage	The two stages
Days past due, both ways	Under each rule set's counting basis
Reason for the difference	The setting that causes it: threshold profile, Mega Farm class, counting basis, maturity count, schedule correction, missed instalments, qualitative trigger or cure
Carrying amount, EAD	As booked
ECL effect	The twelve-month or lifetime ECL at the comparison stage, using the booked PD, LGD and EAD for that stage
RBM class	Unchanged by staging (section 5), shown for reference

A summary gives the count and carrying amount moving between each pair of stages, the ECL effect in total and by reason, and the effect on Stage 3 interest. It is on the Staging & SICR Rules screen, it can be exported, and it is shown on each month's audit workbook.

Uses. It shows each month what adopting directive-aligned staging would change. It reproduces the December 2025 method on the system's data for comparison with the workbooks. And it lets a single setting be tested before it is proposed.

5. Staging and the RBM return stay separate

The RBM return classifies facilities under the directive's own bands: by term, on the count of days from the oldest overdue instalment. It must not change when the staging policy changes. So:

- The loan book build always computes the directive's day count, with the maturity count and the schedule corrections, and keeps it in `overdue_days`. The RBM return and the RBM classification report read only this.
- The days that staging uses are computed under the staging settings (4, 5 and 6) into a separate column, `staging_dpd`, with its basis recorded on the row.
- The comparison of the stage with the RBM class, already in the IFRS 9 vs RBM report, then shows any loan whose stage and class disagree, with the setting that causes it.

6. What follows the staging policy, and what does not

Area	Follows the staging policy?	How
PD by segment and transition matrices	Yes	Measured on the stage movements
Time-phased ECL	Yes	Twelve-month or lifetime by stage
Mega Farm ECL	Yes	Its stage comes from setting 3
EIR revenue	Yes	Stage 3 interest on the net basis
Forward-looking route and scenarios	Yes	Applied to the staged PD
Restructure register	Yes	Records the stage at the event
IFRS 9 note and reports	Yes	By booked stage
RBM return and RBM classification	**No**	Directive day count, always (section 5)
Golden ties of the bootstrap	No	None of the nine reads the stage

7. Data and code

Settings. Settings 1, 3, 5, 6, 8, 9, 11 and 12 are added to the Governance Centre with their options, descriptions and comments; settings 4, 7 and 10 exist and gain the comment and the new seeded value where it changes (10, from 3 to 0; 4 and 7 to the 31 December 2025 options).

Staging service. `StagingService` reads every rule from the settings as at the month-end, with no rule fixed in code and no silent default (decision D21). The loan book build writes `overdue_days` (directive count) and `staging_dpd` with its basis. The maturity count and the schedule correction become switchable in the staging count only; the directive count keeps both always.

Comparison run. A table `staging_comparison_runs` (period, preset or settings snapshot, who ran it, when) and `staging_comparison_rows` (per loan: both stages, both day counts, the reasons, the ECL effect). A command `staging:compare {period} {preset}` and a panel on the Staging & SICR Rules screen.

Audit trail. Every staging run records the settings in force, their effective dates and the preset, so any month can be traced to the policy it ran under.

8. Every Governance Centre change logged, with its supporting documents

Staging becomes governed through the Governance Centre, so the Centre's record has to be good enough for an auditor or the Reserve Bank to follow every policy choice: who proposed it, why, on what evidence, who approved or rejected it, when it took effect and what it replaced. This section covers **every setting in the Governance Centre**, not only the staging ones.

8.1 What is recorded today, and what is missing

Today	Gap
A proposal writes an audit entry ("EIR Governance Setting Proposed") with the old and new values, the effective date and the reason	Only proposals and approvals are recorded. There is no reject or withdraw step: a proposal that is not wanted can only be left standing, with no record of why
An approval writes an audit entry ("EIR Governance Setting Approved"), and the setting it replaces is copied to the history table	The reason is cut to 500 characters on the setting
The proposer cannot approve their own change; an administrator override is itself a governed setting	The seeded defaults are written by the installation with no audit entry
	No document can be attached to a proposal or an approval
	The record is spread across the general Audit Trail screen; there is no single history per setting
	The audit log is an ordinary database table: a database administrator could change or delete an entry without trace

8.2 What every change will record

Each of the following events writes one audit entry, linked to the setting and to the change it belongs to:

Event	Recorded
Seeded	The setting, the seeded value, its effective date, and that it was seeded by installation (not a MAIIC decision)
Proposed	The proposer, the setting, the value in force and the proposed value, the effective date, the reason in full, and the documents attached
Document added or removed	Who, when, the document's name, type, size and fingerprint; a removed document is kept and marked removed, never deleted
Approved	The approver, the time, any comment, the documents attached at approval, and the setting the change replaces
Rejected (new)	The reviewer, the time and the reason (required). The proposal stays on record as rejected
Withdrawn (new)	The proposer, the time and the reason. The proposal stays on record as withdrawn
Preset applied	For a preset (section 3), one entry for the preset and one for each setting it changed, all linked to the same change
Override used	If an administrator approves their own proposal under the governed override, that fact, with the override setting in force

8.3 Supporting documents

- **Where.** A proposal, an approval, a rejection and a withdrawal can each carry documents: for example a board or committee minute, a policy paper, an analysis, the comparison-run export (section 4), or correspondence with the auditors or the Reserve Bank.
- **Required evidence.** Each setting can be marked as needing a document to be approved. The staging presets, `staging_threshold_profile` and `stage2_threshold_medium_long` (a rebuttal under IFRS 9) are marked: adopting directive-aligned staging cannot be approved without the minute or decision that adopts it.
- **Kinds of file.** PDF, Word, Excel, CSV, PNG and JPEG, up to 20 MB each, with a short description and a document type (minute, policy, analysis, correspondence, other).
- **Stored safely.** Files go to the system's **private** storage on MAIIC's server, never to the public web folder, and can be opened only by users with the governance permission. Each file's SHA-256 fingerprint is recorded when it is uploaded and checked when it is opened, so a changed file shows. The existing `supporting_documents` table is reused for this, with its storage set to private.

- **Kept.** Documents are never deleted from a decided change. They are included in the backups (spec v4 and the integration document, storage and backup) and kept for the life of the system.

8.4 One history per setting

The Governance Centre gains a **History** view for each setting: every value it has held, with its effective dates, who proposed and approved or rejected each change, the reasons, and the documents, newest first. The view can be exported to PDF and Excel, and the same history is a sheet in each month's audit workbook.

8.5 A log that cannot be changed quietly

- The governance audit entries are **append-only**. The system has no screen or command to edit or delete them, and the build adds a database guard so that an update or delete on these entries is refused.
- Each entry carries the **fingerprint of the entry before it**, so the entries form a chain. A check (run with the bootstrap's verify step and on demand) recomputes the chain and names the first entry that does not follow. Anyone who changes or removes an entry directly in the database breaks the chain, and the check shows where.

9. Decisions owed

Ref	Decision	Owner	Recommendation
S1	Adopt directive-aligned staging, and from which month	Dr Thom, with the board or its committee	Adopt from the first month not yet locked after the comparison runs have been reviewed; the effect is shown monthly until then
S2	The Stage 2 threshold for medium and long-term facilities (O13)	Dr Thom	Keep 31 days unless documented evidence supports 91
S3	The Mega Farm term class	Finance, then Dr Thom	By tenor, as the directive
S4	The cure test	Dr Thom	Arrears cleared, then 3 month-ends
S5	Whether to keep each accepted departure of section 2.3 visible on the face of the monthly report until S1	Finance	Yes
S6	Which settings require a supporting document before approval, beyond those named in section 8.3	Dr Thom, with Finance	Every setting that is a Phase 0 sign-off, and every staging setting
S7	Who may view the supporting documents	MAIIC ICT, with Dr Thom	Users holding the governance permission, and the auditors through the auditor's pack

10. Acceptance

1. **Seeded default.** On a fresh bootstrap every staging setting holds the 31 December 2025 option, effective from inception, and no directive-aligned setting is in force.
2. **No switch by date.** A month after any date is still staged on the seeded default until a preset is approved in the Governance Centre; the bootstrap never approves one.
3. **Adoption.** Approving directive-aligned staging from a month stages that month and later ones on the new rules; locked months are unchanged.
4. **Reproduction of the December ladder.** Under the 31 December 2025 policy, December 2025 is staged on one ladder (1 to 30 days Stage 1; 31 to 180 Stage 2; over 180 Stage 3) from the bucket ageing, and every difference from the workbooks' 108 facilities is listed with its cause: a facility not in one of the two books, a different arrears bucket, or a different balance.

5 **Comparison.** The comparison run for December 2025 and for September 2026 lists every loan whose stage differs between the two presets, each with its reason and ECL effect, and its totals reconcile to the two stagings.

6 **RBM return unchanged.** The RBM return for a month is identical whichever staging policy is in force.

7 **No silent default.** Removing a staging setting for a month stops staging with a named error, as today.

8 **Regression.** The bootstrap's nine golden ties still pass.

9 **Every change logged.** Seeding, proposing, adding or removing a document, approving, rejecting, withdrawing, applying a preset and using the override each write one audit entry with the fields of section 8.2, and the setting's History view shows them in order.

10 **Required evidence.** A staging preset or a Stage 2 rebuttal cannot be approved without at least one supporting document; the refusal names what is missing.

11 **Documents safe.** An uploaded document is stored outside the public web folder, cannot be opened without the governance permission, and a file altered on disk is reported as altered when opened.

12 **Append-only.** Editing or deleting a governance audit entry through the system is impossible, and doing it directly in the database is refused; a change made with the guard bypassed is found by the chain check, which names the first broken entry.

11. Build order

Step	Work	Size
1	The settings, the presets and their seeding; `StagingService` reading them; `staging_dpd` beside `overdue_days` in the build	About 1.5 days
2	The comparison run: tables, command, panel, export, audit-workbook sheet	About 1 day
3	The governance change log of section 8: reject and withdraw, the seeding entries, supporting documents on private storage with fingerprints, required evidence, the History view and export, the append-only guard and the chain check	About 2 days
4	Tests for each acceptance point; a bootstrap check	About half a day
5	Release to MAIIC's server by the release steps of the integration document (section 2.2): push, pull, migrate, bootstrap with the nine ties passing, then the comparison runs for December 2025 and September 2026	With MAIIC ICT
Later	Setting 9, once each loan's PD at origination is kept	To be planned

12. Glossary

Term	Meaning
Stage 1, 2, 3	IFRS 9's three buckets: performing (twelve-month ECL), a significant increase in credit risk (lifetime ECL), and credit-impaired (lifetime ECL, interest on the net basis)
Days past due	How long the oldest unpaid instalment has been overdue at the month-end
Arrears bucket	A band of overdue amounts in E-Banker's loan book report (1 to 30, 31 to 90, 91 to 180, 181 to 270, 271 to 360 days)
Single-ladder staging	MAIIC's staging policy as at 31 December 2025: one set of day thresholds for every facility, read from the arrears buckets
Directive-aligned staging	Day thresholds by facility term under the RBM DFI Directive 2018, with MAIIC's missed-instalment policy, qualitative triggers, data corrections and a cure test
Preset	A named set of all the staging settings, chosen in one governed change
Comparison run	A staging of one month under another preset or set of settings, beside the booked staging, that changes nothing booked
Cure period	The month-ends a loan must stay below a threshold before it moves to a lower stage
Rebuttal	Setting aside an IFRS 9 presumption (30 days for Stage 2, 90 days for default) on documented, reasonable and supportable evidence
Maker-checker	One person proposes a change and a second person approves it
Supporting document	A file attached to a governance change as its evidence: a minute, a policy paper, an analysis or correspondence
Append-only	A record that can be added to but never changed or deleted
Chain check	A test that each audit entry still carries the fingerprint of the one before it, so that a changed or removed entry is found